

What AI Share-of-Voice Doesn't Capture

**A pre-registered convergent-validity test of AIAS Presence
against a third-party AI-visibility instrument**

May 2026 · Pablo Ulpiano González Castro

AIAS™ Measurement Program · Convergent Validity

AIAS Presence does not agree with the “Share of Voice” score that commercial AI-visibility tools put at the top of the dashboard. Across 24 B2B SaaS brands, the correlation was weak and not significant — a pre-registered null. But the same measure converges strongly with the tool’s presence-quality and brand-recognition signals. The headline number is the weakest dial on the board; the signals beneath it move with AI presence.

rho = 0.29 Primary convergence ·
FALSIFIED Pre-registered hypothesis ·
0.80 / 0.75 Where it does converge · **6 / 6**
 Recognition at ceiling

This study tested whether AIAS Presence agrees with the score that commercial “AI visibility” tools sell as a measure of how present a brand is inside ChatGPT, Perplexity, and Gemini. It does not. Across 24 B2B SaaS brands, the rank correlation between AIAS recall-SOM and the HubSpot AEO Grader’s Share-of-Voice was $\rho = 0.29$ — not significant, with a confidence interval that straddles zero. Share-of-Voice was the metric named in advance as the convergent variable, and the test failed.

The null is not the whole story. An exploratory scan of the instrument’s other dimensions found that AIAS recall-SOM converges **strongly** with two of them — presence quality ($\rho = 0.80$) and brand recognition ($\rho = 0.75$), both above the program’s 0.74 “strong” benchmark — while the headline Share-of-Voice it was tested against carried almost no rank information. These correlations are exploratory; the pre-registered metric failed, and re-nominating a

better-correlating dimension after the fact is not available. But the pattern is informative.

The failure is in the metric, not the construct. It is tempting to read the Share-of-Voice null as proof that AIAS Presence measures something category-competitive that brand-absolute tools cannot see. The data refute that: presence quality and brand recognition are also brand-absolute scores, and recall-SOM converges with them. Nor is it range restriction — Share-of-Voice actually varies more than the dimensions that beat it. The problem is that its variation does not co-rank with AIAS recall: it is tie-heavy (7 of 19 live brands sit at one value), and it measures competitive share, while the dimensions that do converge — presence quality, brand recognition — measure brand knowledge and prominence, the same thing recall captures. They vary less, yet they track recall, consistent with AIAS’s earlier convergence with Google Trends search interest (v0.25, $\rho = 0.74$).

For brand leaders, the consequence is sharper than a clean convergence would have delivered: the “Share of Voice” headline these tools sell as AI visibility is, on this evidence, the weakest available proxy for category-competitive AI presence. A brand managing to that number may be watching the least

informative dial on the dashboard.

What we measured

We correlated AIAS Presence against a commercial AI-visibility instrument across 24 B2B SaaS brands, pre-registered before any instrument data was pulled.

AIAS-side (recall-SOM): A brand's share of category-leadership recall across a six-model panel — how often each brand is named when a model is asked, generically, for the leading platforms in its category (0–100). The measure is inherited verbatim from the v0.25 scorer, so it is identical to the one that converged with Google Trends at $\rho = 0.74$. Brand recognition (C_p) sat at the ceiling — every brand scored 6/6 — so it carries no variance and enters only as a null control.

Instrument (HubSpot AEO Grader, free tier): For each brand, the Grader returns a perception profile across three engines on five dimensions — Share of Voice, Presence Quality, Brand Recognition, Market Competition, and Brand Sentiment — plus an overall composite. **Share of Voice** was pre-registered as the convergent variable.

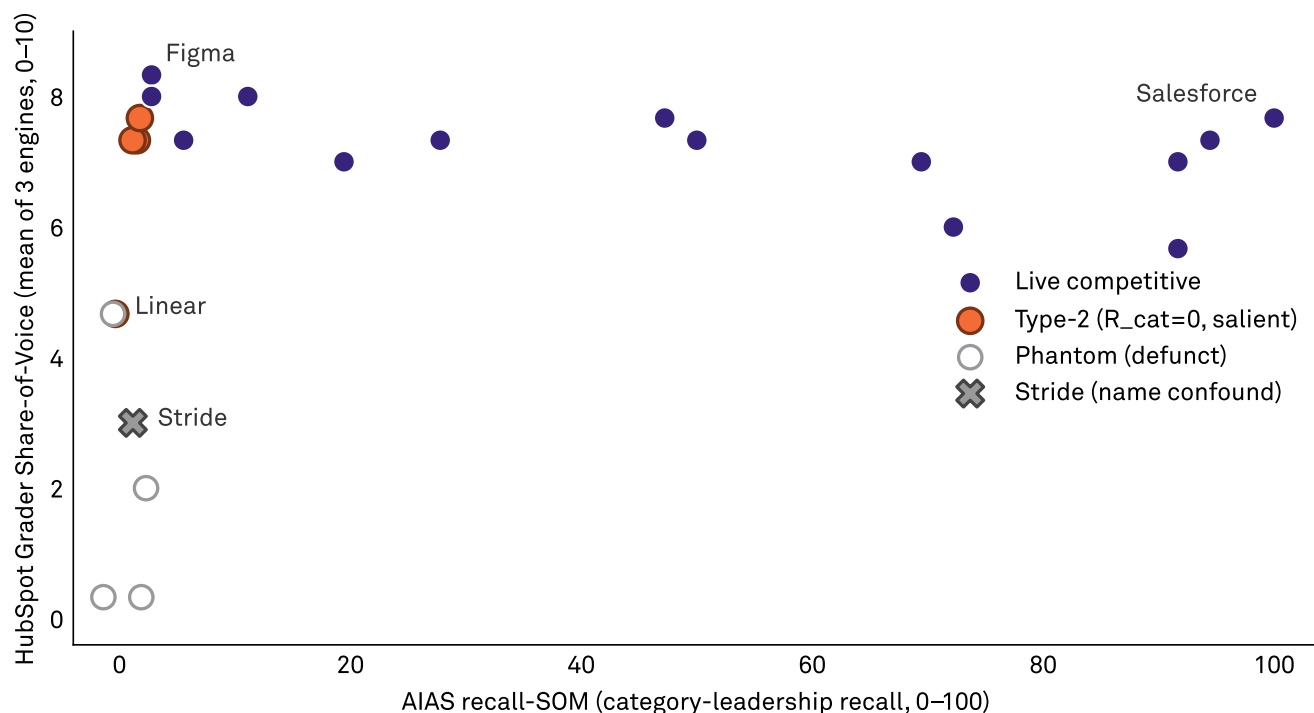
Test: Spearman rank correlation (10,000-sample bootstrap, Holm correction), with two pre-specified sensitivity analyses. The construct-matched instrument (Profound's category-scoped share-of-model) was deferred on access; a discriminant instrument (Brandwatch) was not run. The convergent claim here rests on the HubSpot Grader alone.

FINDING 01

The headline metric doesn't converge

AIAS recall-SOM vs. brand-absolute AI Share-of-Voice

HubSpot AEO Grader Share-of-Voice, mean of 3 engines (n = 24)

Spearman $\rho = 0.29$ (n.s.); much rank agreement is the live/defunct split.*Primary convergent test falsified — recall-SOM does not converge with brand-absolute Share-of-Voice.*

Source: AIAS™ v0.27 | Protocol v1.6 | Third System™

Figure 1 · AIAS recall-SOM does not converge with the Grader's Share-of-Voice. Category-leadership recall (0–100) vs. HubSpot Share-of-Voice (0–10) across 24 B2B SaaS brands; $\rho = 0.29$ (n.s.). Much of the rank agreement is the live-versus-defunct split. Type-2 brands (specialist-salient, zero category recall) sit at the left edge with mid-to-high Share-of-Voice; Stride carries a name-collision confound.

AIAS recall-SOM and the HubSpot Grader's Share-of-Voice do not co-rank: $\rho = 0.29$ across 24 brands (not significant; 95% CI [-0.17, 0.64]). The correlation sits below the 0.60 confirm threshold and is statistically indistinguishable from zero. The pre-registered primary test is falsified.

Much of the rank agreement that does exist is carried by the live-versus-defunct split — both measures rank the dead products (Wunderlist, HipChat, Yammer) at the bottom. Above that floor, the Grader's live-brand scores barely move: Salesforce, Figma, Stripe, and Cloudflare all land within a point of each other on Share of Voice while their

category-leadership recall spans the full range. A brand can post a strong Share-of-Voice score and be absent from the category-leadership answers its buyers actually receive — and the reverse.

The clearest cases are the four “type-2” brands (Linear, Airtable, Miro, Cloudflare): specialist-salient, visible when the Grader scores them in isolation, but absent from generic category-leadership recall. A pre-specified sensitivity that drops them lifts the correlation only to $\rho = 0.36$ — still below threshold. The non-convergence is broader than those four cases.

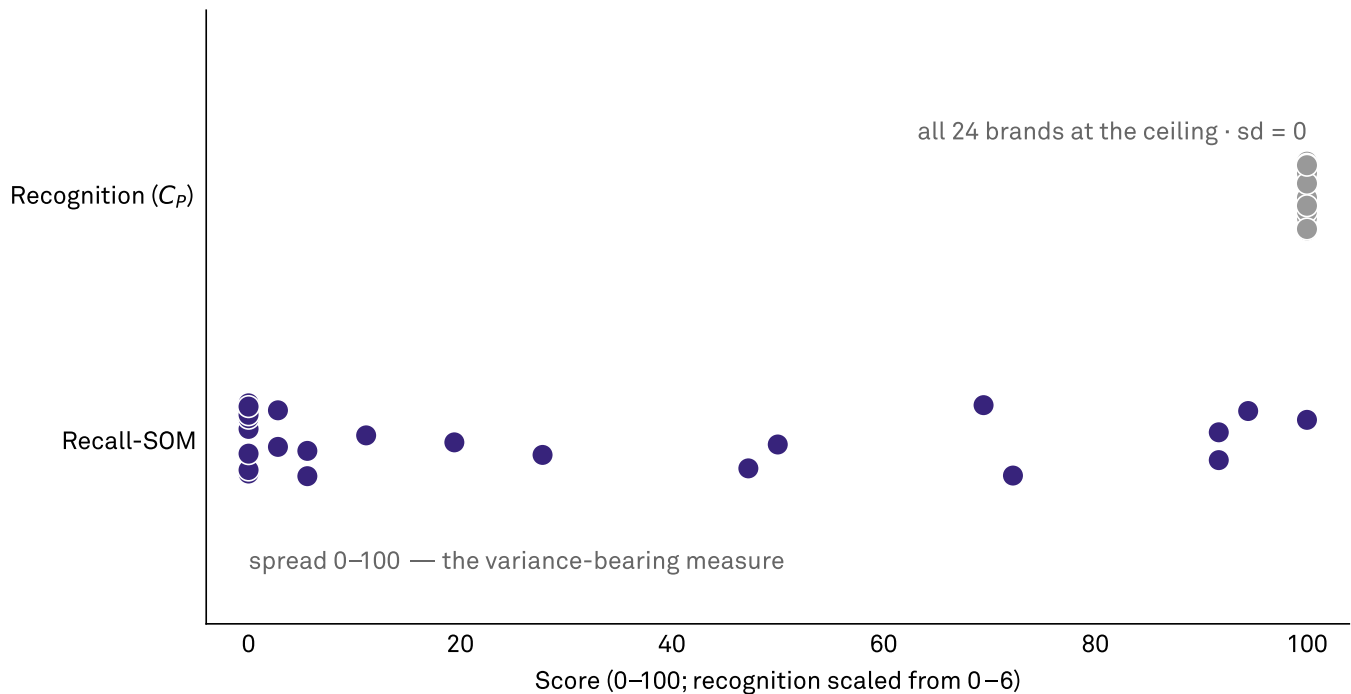
FINDING 02

Recognition is already maxed out

Recognition is at the ceiling; recall carries the variance

AIAS recognition (C_p) vs. recall-SOM across 24 B2B SaaS brands

Recognition is saturated at 6/6 for every brand; only recall-SOM separates them.



Recognition enters as a null control; recall-SOM is the variance-bearing convergent variable.

Source: AIAS™ v0.27 | Protocol v1.6 | Third System™

Figure 2 · Recognition is at the ceiling; recall carries the variance. AIAS recognition (C_p , scaled 0–100) vs. recall-SOM across 24 B2B SaaS brands. Every brand scores 6/6 on recognition ($sd = 0$), so it cannot rank brands; only recall-SOM separates them. Recognition therefore enters as a null control, not a convergent variable.

Every one of the 24 brands scored full recognition ($C_p = 6/6$) across the model panel. With zero variance, recognition carries no convergent signal by construction — it cannot rank brands because it does not separate them.

This is why AIAS Presence is operationalized as recall-SOM rather than recognition on this substrate: in a category where

models recognize every established brand, the question that discriminates is not “does the model know this brand?” but “does the model surface it when asked for the category leaders?” The ceiling is a property of B2B SaaS, recorded here so recognition is not mistaken for a live convergent dimension.

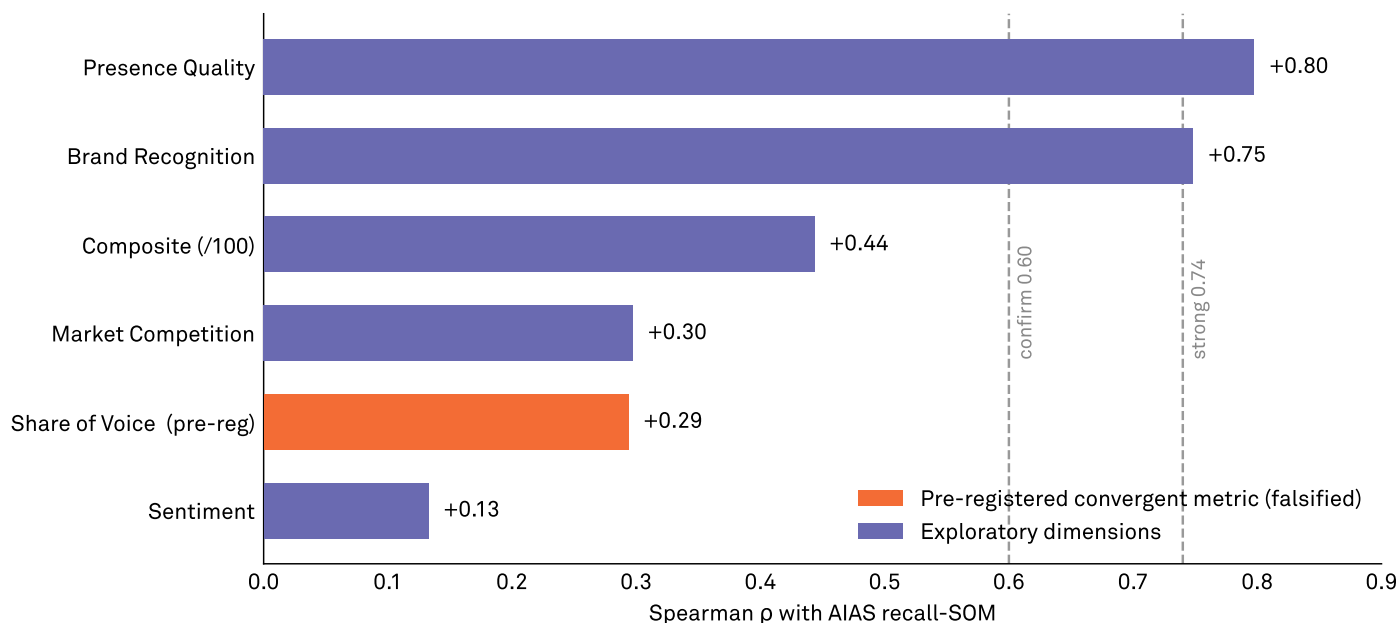
FINDING 03

Where AIAS Presence does converge

Where AIAS recall-SOM meets the HubSpot Grader's dimensions

Spearman ρ of recall-SOM with each Grader dimension (n = 24)

Exploratory; the pre-registered metric is Share of Voice.



Presence-quality and brand-recognition exceed the 0.74 strong benchmark; SoV does not.

Source: AIAS™ v0.27 | Protocol v1.6 | Third System™

Figure 3 · Where AIAS recall-SOM meets the Grader's dimensions. Spearman rho of recall-SOM against each HubSpot Grader dimension (exploratory, n = 24). The pre-registered metric, Share of Voice, is among the weakest; presence quality and brand recognition exceed the 0.74 "strong" benchmark. Exploratory — does not bear on the falsified primary.

The pre-registered metric failed, but the instrument has five other dimensions, and AIAS recall-SOM tracks two of them closely. Against **Presence Quality** the correlation is $\rho = 0.80$; against **Brand Recognition** it is $\rho = 0.75$ — both above the program's 0.74 "strong" benchmark. The overall composite reaches $\rho = 0.44$. Share of Voice, the dimension it was tested against, is the weakest match at $\rho = 0.29$.

These correlations are **exploratory**. Share of Voice was the pre-registered convergent variable; it failed, and re-nominating presence quality or brand recognition as the

"real" convergent measure after seeing which one worked is exactly the move pre-registration exists to prevent. They are reported to generate the next hypothesis, not to rescue this one — and they are pre-specified for confirmatory test on a fresh substrate.

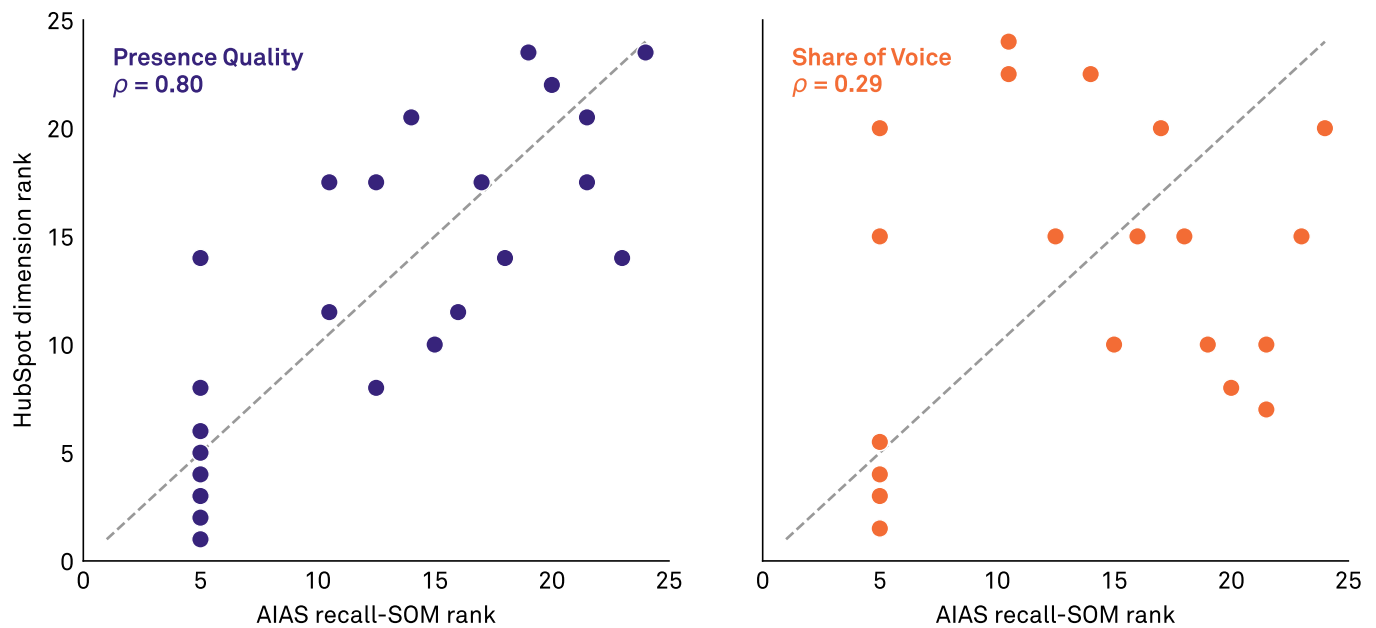
What they show is that AIAS recall-SOM is not failing to converge with this instrument in general. It converges with most of the instrument's facets and fails on the one the vendor puts in the headline.

FINDING 04

It's the metric, not the construct

Presence quality's ranking follows recall's; Share of Voice's doesn't

Rank–rank plots: AIAS recall-SOM rank vs. HubSpot dimension rank (n = 24)

Dashed line = perfect rank agreement. Presence quality hugs it; Share of Voice scatters — its variation is real but rank-misaligned with recall.*The convergence gap is rank-alignment, not spread: Share of Voice varies as much, but its order does not match recall's.*

Source: AIAS™ v0.27 | Protocol v1.6 | Third System™

Figure 4 · Rank-alignment, not spread. Rank–rank plots of AIAS recall-SOM rank vs. HubSpot dimension rank (n = 24). Presence quality hugs the perfect-agreement diagonal ($\rho = 0.80$); Share of Voice scatters off it ($\rho = 0.29$). Share of Voice varies as much, but its order does not match recall's — and it is tie-heavy (7 live brands at 7.33/10).

The obvious reading — that AIAS Presence measures category-competitive presence and therefore diverges from brand-absolute AI visibility — does not survive the data. Presence quality and brand recognition are computed on a single named brand in isolation, exactly like Share of Voice, and recall-SOM converges with them. The category-versus-brand-absolute story does not explain the pattern.

What does explain it is rank-alignment, not spread. Share of Voice actually varies more than the dimensions that beat it — its spread exceeds presence quality's and brand recognition's — but its variation does not co-rank with recall. Two things drive the gap: Share of Voice is tie-heavy (7 of 19 live brands sit at exactly 7.33/10, so it can barely order salient brands), and it measures competitive share rather than the brand-knowledge prominence recall captures. Presence quality and brand

recognition vary less, yet their order tracks recall. One detail sharpens this: the Grader's own brand-recognition subscore tracks recall-SOM even though AIAS's recognition measure is at the ceiling — the graded version behaves like a prominence measure, which is what recall-SOM is.

Read against the program's prior results, the pattern is coherent. The same AIAS measure converged with Google Trends search-interest prominence ($\rho = 0.74$) and stood independent of Amazon sales rank. The dimensions it converges with here — prominence and knowledge facets — are of a piece with the Trends result; the rank-misaligned competitive-share score it fails on is the exception the construct predicts. The practitioner takeaway is direct: the "Share of Voice" number sold as AI visibility carries real signal, but read as AI availability it is the wrong place to read it.

Limitations

The convergent claim rests on a single instrument, pre-registered as a **loose** construct match. The strongest available test — convergence against a category-scoped, construct-matched instrument (share-of-model) — was never run: Profound is enterprise-only and access was declined as disproportionate to one phase. A single-instrument null on one dimension cannot, on its own, separate a genuine convergence failure from a metric artifact — which is what the exploratory scan suggests is in play.

The Grader is brand-absolute and cannot be category-scoped on the free tier. One brand, Stride, carries a name-collision confound: “Stride” is ambiguous (Stride gum, Stride Inc., Stride Bank), so its score may reflect non-target entities rather than the defunct Atlassian product. A pre-specified sensitivity excludes it; the estimate is **sensitive** to that exclusion, which is reported rather than smoothed.

$n = 24$ is modest, though pre-registered and adequate for the planned correlation. Recognition is at the ceiling on this substrate, so recognition convergence cannot be tested here. The AIAS-side measure is inherited from the v0.24 corpus, collected the same month as the instrument pull; residual brand-level drift over the gap is mitigated, not removed, by rank-based correlation.

What's next

The deferred test is the decisive next step: convergence of AIAS recall-SOM against a category-scoped share-of-model instrument (Profound, or a self-serve tool with equivalent scoping). If recall-SOM converges with a construct-matched instrument, the present null becomes evidence of construct specificity rather than a threat to it.

Separately, the strong exploratory convergence surfaced here — presence quality and brand recognition — is pre-specified for a confirmatory pre-registration on a fresh substrate, with those dimensions fixed in advance, to test whether the pattern replicates. Completing the discriminant arm (a web-mention instrument) would round out the Campbell–Fiske matrix paired with the v0.26 sales-rank result.

Hypothesis scoring

Four pre-registered propositions for the v0.27 convergent-validity test of AIAS recall-SOM against the HubSpot AEO Grader (B2B SaaS, n = 24).

H	Pre-registered prediction	Result	Status
P1	Primary convergence — recall-SOM × Share-of-Voice	$\rho = 0.29$, $n = 24$, n.s. (95% CI [-0.17, 0.64]). Below threshold.	FALSIFIED
P2	Recognition null control	$C_p = 6/6$ for all 24 brands ($sd = 0$). No variance, as predicted.	CONFIRMED
P3	Component scan — where AIAS Presence converges	Presence Quality $\rho = 0.80$, Brand Recognition $\rho = 0.75$ (both > 0.74); Composite 0.44; Share of Voice 0.29 (weakest).	EXPLORATORY
P4	Construct-matched instrument (share-of-model)	Profound access declined as disproportionate (enterprise-only). Deferred.	NOT RUN

Detailed hypothesis results

P1 — Primary convergence: FALSIFIED recall-SOM × HubSpot Share-of-Voice: $\rho = 0.294$, $n = 24$, $p_{adj} = 0.164$ (Holm), 95% CI [-0.171, 0.638]. Below the 0.60 confirm threshold and not significant. Type-2 sensitivity (drop Linear, Airtable, Miro, Cloudflare): $\rho = 0.359$ ($n = 20$). Stride-confound sensitivity (drop Stride): $\rho = 0.223$ ($n = 23$) — the estimate is sensitive to the exclusion.

P2 — Recognition null control: CONFIRMED Brand recognition $C_p = 6/6$ for all 24 brands ($sd = 0$). Zero variance; recognition enters only as a control, not a convergent dimension.

P3 — Component scan: EXPLORATORY recall-SOM × Grader dimensions: Presence Quality $\rho = 0.798$ ($p < 0.001$), Brand Recognition $\rho = 0.748$ ($p < 0.001$), Composite $\rho = 0.443$ ($p = 0.030$), Market Competition $\rho = 0.297$, Share of Voice $\rho = 0.294$ (n.s.), Sentiment $\rho = 0.133$. Exploratory; does not bear on the falsified primary.

P4 — Construct-matched instrument: NOT RUN Profound (category-scoped share-of-model) deferred on access at pre-registration amendment r4; the discriminant instrument (Brandwatch) was not run.

About the Third System

The Third System describes a shift in how consumers make decisions: instead of browsing and comparing options directly, many now rely on AI systems to generate recommendations. In this environment, brands compete not only for human attention but for inclusion within AI-generated outputs. This creates a new competitive layer where visibility is determined by how AI systems interpret and surface brands.

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Methodology

This report presents findings from Third System's AI Availability research program. The AI Availability Score (AIAS) is an early-stage measurement framework introduced in Tri-System Brand Growth (Gonzalez Castro, 2026). Findings should be used directionally. Construct validity — the correlation between AI Availability and consumer behavior — has not yet been established and is the subject of Phase 3 of the research program. Methodology is published in full and versioned at thirdsystem.ai/methodology (current: v1.6). Registry construction is curated and reflects the framework's view of each category's competitive set; revisions are documented in the methodology log.

Citation

González Castro, P. U. (2026). *A Pre-Registered Null on AI Share-of-Voice: Convergent Validity of AIAS Presence against Third-Party AI-Visibility Instruments (v0.27)*. Third System. thirdsystem.ai/v27

Underlying datasets

AIAS-side: osf.io/ec6wh/v27/data/v0.27_aias_side.csv

Instrument I1: osf.io/ec6wh/v27/data/v0.27_I1_normalized.csv

Scoring verdicts: osf.io/ec6wh/v27/v0.27_verdicts.json

Methodology log: v1.6 (SSRN 6816340).

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